

F24-048-D-unHinged

Project Team

Arham Malik	21I-0441
Tuba Khan	21I-0861
Saad Jamil	18I-0437

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Supervised by

Ms. Nirmal Tariq



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

Gaming Industry is one of the largest industries in the world along with a huge platform. Among Gen Z 90% identify as gamers whereas in Generation Alpha, the newest generation, 94% identify as gamers[1]. Keeping this huge platform in mind, we want to develop a game that offers entertainment but at the same time it offers a new perspective, a perspective of understanding the mentally challenged population by portraying how a handful of mental illnesses can disturb and disrupt your daily life functionalities. The mental illnesses that will be focused in our game are Schizophrenia, Nyctophobia, and the brain disorder Alzheimer's. Through the replication of these mental illnesses and use of procedural generation, we might be able to spread some awareness about mental health as well as players' interest in the game. Procedural generation, while not a new concept, has gained attraction in game development. Early examples were limited, but recent advancements have enabled its wider application. [2].

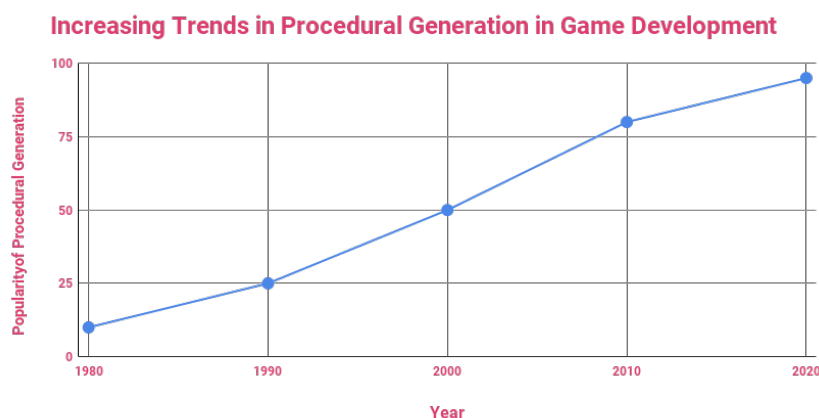


Figure 1.1: Increasing Trends

1.1 Problem Statement

Mental Health is a topic that has gained much attention in recent years and is being taken much more seriously. To coincide with this the portrayal of mental health in media is observed with a higher degree scrutiny to prevent any negative stereotypes from developing. Our team hopes to show the struggles of those with mental health issues and brain disorders while also creating an experience that can be enjoyed by consumers. We also would like to highlight the potential of procedural generation in game development. The use of procedural generation would allow developers to create quality products while reducing the workload and the costs needed. This is due to the increasing demands of quality and scope of games in recent years which procedural generation is one solution to meet these demands.

1.2 Scope

The scope of our project focuses on creating a procedurally generated video game that blends entertainment with raising mental health awareness. The game offers players an immersive experience where they will face challenges from the perspective of someone dealing with mental health issues. Each game level will be designed with guidance from mental health professionals to simulate the real-world difficulties faced by individuals with these conditions. The game will be a 2D, top-down, single-player experience, aiming to promote social awareness while delivering engaging gameplay. Set in a survival environment, the player's goal is to escape a mental asylum, but without any violent or aggressive content, this is to combat the stereotype of mental illness leading to violent behaviors. The game's adaptive difficulty will cater to different skill levels, allowing players of all abilities to enjoy the experience while contributing to the message of empathy and understanding.

1.3 Modules

The project will be divided into the following four modules:

1.3.1 Game Story & Prototypes

This module is responsible for setting us on a path of Game Development while starting with the documentation and getting started with the project while using appropriate software.

1. **Game Design Document:** Creation of Game Design Document.
2. **Prototype:** Medium Fidelity Prototype on Figma
3. **Setup:** Installation and Setting up Blender & Godot
4. **Backstory:** Creating a game backstory to develop a game around while also generating necessary dialogues for level 1.

1.3.2 Basic Game Structure

Module 2 is responsible for having basic assets created and adding basic functionality of the game while overcoming the challenges that arise during the initial phase.

1. **Asset Creation:** Creation of 2D Assets to be used in the game.
2. **Asset Animation:** Animating the 2D Assets in accordance with game.
3. **Game Menu:** Creation of Game Menu and Settings.
4. **Basic Level Implementation:** Setting the initial Level 1.
5. **Tutorial:** adding a tutorial in Level 1
6. **Torch Mechanism:** Adding Torch with ON and OFF Mechanism and adjusting lightning along with shaders.
7. **Basic Procedural Generation:** Controlling the randomness of level and optimising its consistency

1.3.3 Advanced Game Structure

Module 3 is responsible for adding Non-Playable Characters (NPC) and their moves along with optimization of the processes in the prior module.

1. **Implementation of Enemy NPC:** Adding Enemy Detection & Chasing Functionality.
2. **Implanting Mental Illness Mechanism:** Adding challenges, hallucinations effect, hallucination meter, and fear meter
3. **Collectibles and Inventory System:** Spreading collectibles through out the level and adding inventory system.
4. **Sound Feature:** Working on Game Sounds

1.3.4 Test & Publishing

It is the final module of the game where we will optimize all the prior issues and add extra functionalities if need arises.

1. **Testing:** Testing the game by playing it over and over again looking for bugs to debug.
2. **Publishing:** Game Deployment on Itch.io, Isotopic, Game Jolt, Kongregate and New Grounds as they offer us to publish games for free.
3. **Marketing:** Marketing the game so that people can start playing the game and bring it to top.

1.3.5 Optional Features

It is the final module of the game where we will optimize all the prior issues and add extra functionalities if need arises.

1. **Adding Backstory:** Comic based story-line to add the backstory to a game in an immersive manner.
2. **Side Quests:** Adding friendly NPC's for side quests.
3. **Speech Patterns in the Game:** Adding Language Modules (LM) and Speech so that user can always hear what the NPCs are talking about instead of reading the conversation.
4. **Adding Choices and Multiple Endings:** Adding multiple endings based on few key choices.
5. **Hidden Paths:** Adding hidden paths throughout the games to reach the exit as fast as possible by going through more difficult hurdles.

1.4 User Classes and Characteristics

Mobile Players	Players that seek a game that can be played in short bursts and run on mobile systems.
Explorers	A type of player that wants to explore the in-game environment to find secrets or easter eggs.
Achievers	A type of player that wants to earn achievements, collectables or simple bragging rights.

Table 1.1: User Classes

Chapter 2

Project Requirements

2.1 Storyboarding & Design Document

See the Figma submission for storyboard: [Figma](#)

See the Game Design Document: [Game Design Document](#)

2.2 Functional Requirements

2.2.1 Module 1: Game Story & Prototypes:

1. The Game Design Document will detail the blueprints needed to complete the project
2. The Figma prototype will show how the first few screens of game may look
3. Game will be designed using Godot
4. Assets will be created using Blender
5. A backstory that ties into the themes of the game

2.2.2 Module 2: Basic Game Structure

1. The game must use 2D assets
2. The Game will feature animations using 2D assets
3. The game will feature a main menu

4. The player can start a new game
5. the player can continue an old game
6. The game will have adjustable settings
7. Level 1 will be created
8. The game will have a tutorial
9. The player can turn the torch on and off
10. The level will be procedurally generated

2.2.3 Module 3: Advanced Game Structure

1. The Game will have NPCs
2. Enemy NPCs will chase the player
3. The player will experience different effects based on mental parameters
4. The game will have an inventory
5. The game will have items to collect and use
6. The game must have sound effects for the atmosphere
7. The game must have 3 levels

2.2.4 Module 4: Test & Publishing

1. The game must be completable
2. The game will be published online for free
3. The game will be marketed to players

2.3 Non-Functional Requirements

2.3.1 Performance

The Game aims to run smoothly in order to not disrupt the tone or frustrate the player. The game will aim for fast load times and stable frame rates to maintain quality. This is especially important for the procedural generation to seen seamless.

PER-1: The game should run at a stable 30 - 60 fps

PER-2: The load times between levels should be no more than 5 seconds long

2.3.2 Atmosphere

The game seeks to convey the feelings and troubles a person with mental health issues goes through and the bleak setting of an asylum. In order to immerse the player in the game, the tone and atmosphere must remain consistent. This will be achieved using the assets created and the sound effects. These features must be coordinated to convey the oppressive and dark setting of an asylum.

2.3.3 Replayability

The aim of procedural generation is not only to lessen the workload of developers and reduce costs, but to also provide more entertainment. Since the levels are created at run time, the player will always get a different experience every time they run the game. This allows a game's lifetime to increase since completing the levels does not mean the player has seen all there is to see. Every time the player starts a new game or level, the layout of the level, challenges and objectives will be slightly altered.

2.3.4 Portable

We are aiming for low end gaming and want our game to reach the broadest audience possible. To achieve this the game must be able to run on any system regardless of processing power. The use of 2D assets will help achieve this, since they are less process heavy than 3D graphics. Godot's lightweight nature also means it will be easier for older and weaker systems to run the game.

2.4 Domain Model

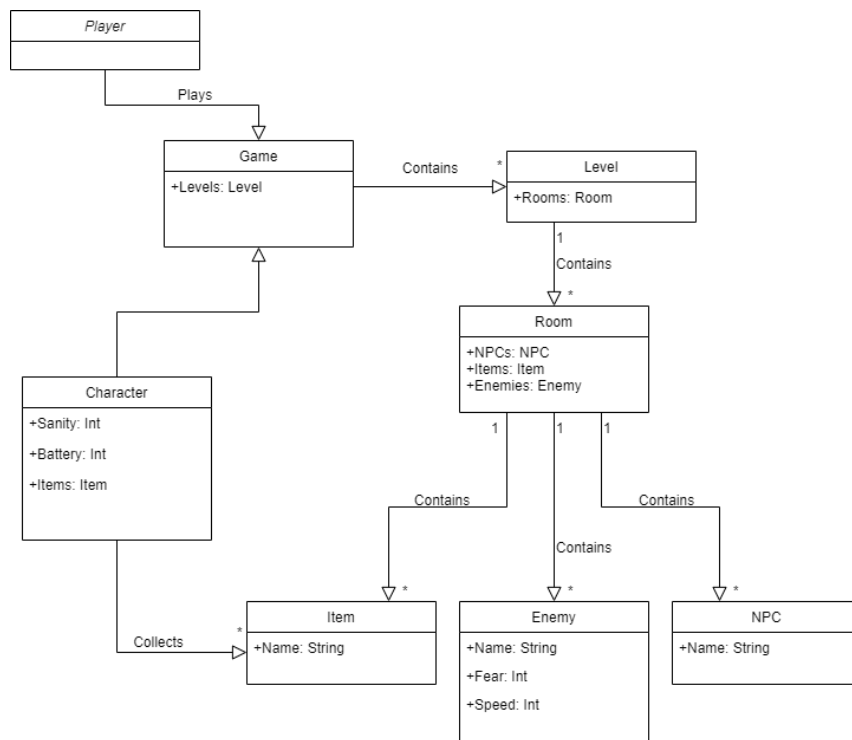


Figure 2.1: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

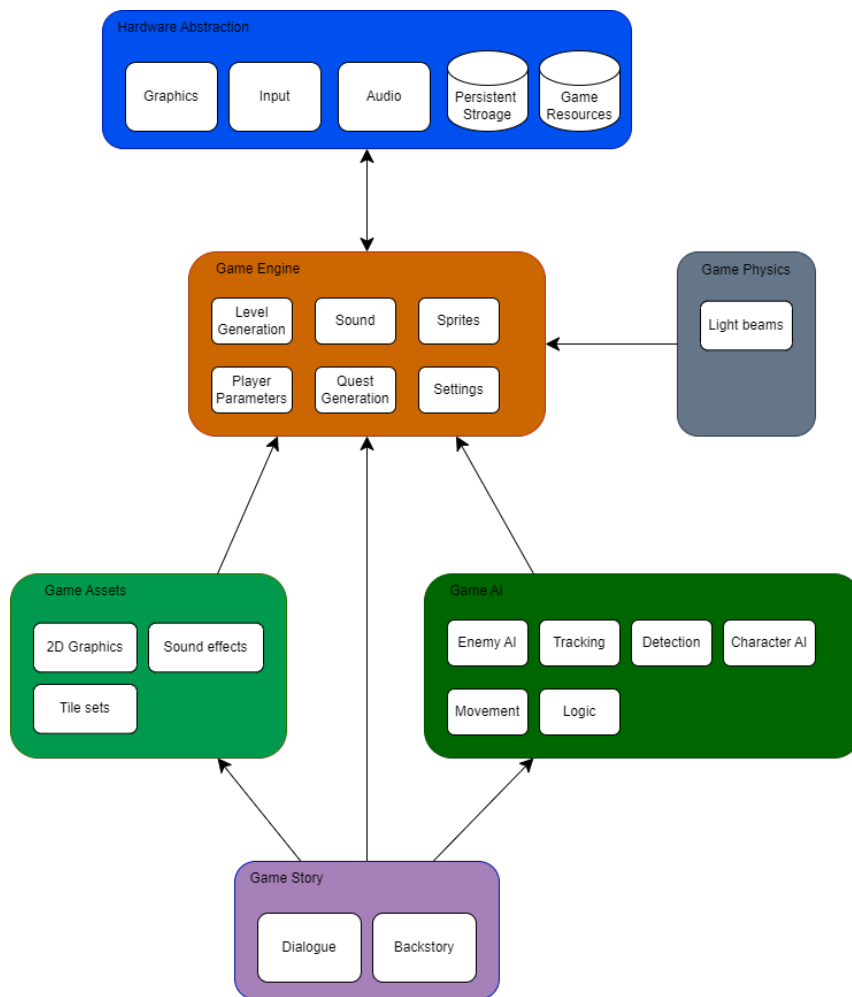


Figure 3.1: Architecture Diagram

The project will be using the architecture as shown above. The architecture is component based, each module is focused on certain functional requirements that need to be met.

- The main module is the Game engine which will be handling the major functions of the game. It will be interacting with all the other modules in order to achieve this.
- The Game Assets will comprise the graphics and sounds the the engine will use to create the game world and levels.
- The Game physics will be the light that the player controls to see around the level and will interact with enemy AI.
- The Game AI handles the behaviours of the entities within the levels and how they respond to the player's actions.
- The story will dictate how each of the levels will be generated and using which assets. The design of the assets will also be influenced by the narrative and setting. Enemy AI will affect how the dialogue will play out, and the story in turn will determine the enemy behaviour
- The Hardware Abstraction will be used to store the state of the game between sessions and will handle the storage of assets. The input provided will tell the engine how to move the player and what the AI should react to.

3.2 Design Models

Procedural Approach

Our project will follow a procedural methodology as this will fit the nature of how the game will run. The game will constantly check the states of certain values and run the needed processes.

The flow chart showcases the runtime processes of the game. Although this is a simplification of the actual flow of the game, the basic structure is similar. The game starts and the player chooses to either load an old game or start a new one. If a new game is started, a new level is generated along with NPCs and enemies. If an old game is loaded, no generation needs to take place and the old state is loaded. The player is then spawns into the game world and what follows depends on the player's input. If the player inputs escape, the game will pause and the option to save, exit or unpause are offered. If the player inputs a direction, then the engine will update the player's position on in the world. When the player moves, the engine will also check to see if an enemy has spotted the player, if so the enemy AI will begin the chase. This flow will follow a cycle until the player reaches the exit, loses all sanity or pauses and exits.

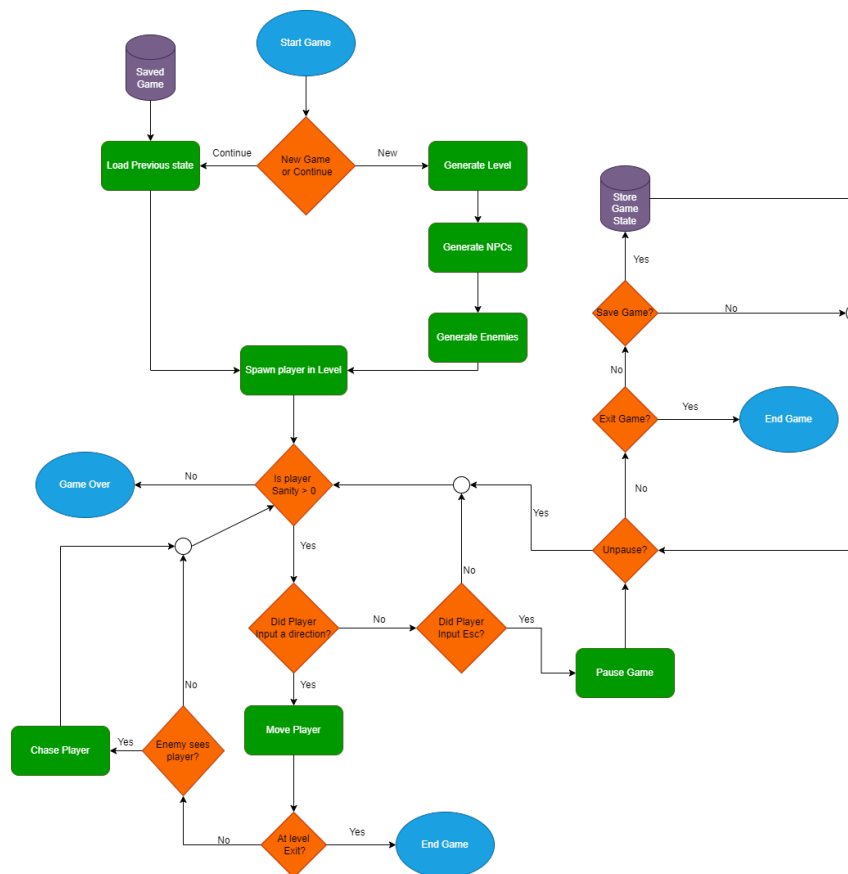


Figure 3.2: Flow Chart

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[2] Tarn Addams Tanya X. Short. Procedural Generation in Game Design. Taylor & Francis, CRC Press, 2017, 2017